
Setup:

Step1: Linux SetUp

Command:{sudo apt update}

 { sudo apt upgrade }

Step2: Python installation

Command:

 { sudo apt install python }

 { python3 --version }

Step3: Install pip:

Command:

 { sudo apt install python3-pip }

Or { sudo apt install python3 python3-pip }

Step4: install Virtual Environment

Command:

 { sudo apt install python3-venv } => install

 { python3 -m venv ~/ai_env } => create vir env

 { source ~/ai_env/bin/activate } => activate

Step4: install PyTorch

Command:

{ ~/ai_env/bin/pip install torch torchvision torchaudio } => *only this works error free*

~~{ ~/ai_env/bin/pip install torch torchvision torchaudio --index-url
<https://download.pytorch.org/whl/cpu> }~~

OR,

~~{ pip install torch torchvision torchaudio --index-url
<https://download.pytorch.org/whl/cpu> }~~

Version check: { pip shows torch }

Step6 : Install Tensorflow/keras/Numpy/Pandas/Matplotlib/Tensorboard/ Seaborn

⇒ ~~Use path before installing locally~~

⇒ **Use inside pip (~/ai_env/pip)**

Command: { ~/ai_env/bin/pip install tensorflow } or { pip install tensorflow }

{ ~/ai_env/bin/pip install keras }

{ ~/ai_env/bin/pip install numpy }

```
{ ~/ai_env/bin/pip install pandas }
```

```
{ ~/ai_env/bin/pip install matplotlib }
```

```
{ ~/ai_env/bin/pip install tensorboard }
```

```
{ ~/ai_env/bin/pip install seaborn }
```

Step7: Install jupyter

Command:

```
{ ~/ai_env/bin/pip install jupyter }
```

.....
Using Jupyter Notebook on a remote server from your PC
.....

Step1: ON PC, open CMD window and open an SSH connection with the server using the following command:

Command: *(1st on Server terminal and PC terminal)*

```
ssh -L <local_port>:localhost:<remote_port> <remote_user>@<remote_host>
```

eg: *created*

```
{ ssh -L 'localhost' admin@(ip address) }
```

[where

0000 is the local port on the PC you selected,

1111 is the port for the Jupyter Server,

admin is the user name for the server,

and @11.11.11.11 is the ip address of the server]

Step 2: ON PC Once you are logged in Server, activate the environment

Command:

```
{ source ai_env/bin/activate }
```

Step3: ON Server/Tera Term;

To open Notebook on PC, Jupyter Server on the remote server This has to be within the environment. Use the following command. make sure that the port is the

Command:

```
{ jupyter notebook --no-browser --port=0000}
```

Step4: On your PC, open a browser window and paste the URL from the step-3 with the token.

{ link of localhost}

You must keep te SSH connection active while you are using the jupyter notebook

*[Note that you have to change the port from
0000 to the localport on the PC to 1111 as
shown above in step 1]*

Step5: Open Note book and Practice on web

.....
New VIRTUAL ENV
.....

Step1: Activate old virtual environment and freeze environment there

Command:

```
{ source ai_env/bin/activate }
```

```
{ ~/ai_env/bin/pip freeze > ~/ai_env/bin/requirements.txt } => freeze
```

Step2: deactivate old environment

Command:

```
{ deactivate }
```

Step3: New Virtual environment create and activate

Command:

```
{ python3 -m venv ~/kustum } => create vir env
```

```
{ source ~/kustum/bin/activate } => activate
```

Step4: install the packages

Command:

```
{ ~/ai_env/bin/pip install -r ~/ai_env/bin/requirements.txt }
```

Step5: Go to this Server connection process

.....
[Follow steps from Using Jupyter Notebook on a remote server from your PC](#)
.....

.....

New User and Virtual Environment

.....

Work in Admin and create new User, first work in main admin to provide access to New user;

Step1: *Create new username out of admin*

Command:

```
{ sudo adduser student }
```

Step2: *login to Studen user with Pw*

Command:

```
{ su student }
```

Step 2: Create New Environment inside Student user

Command:

```
{ python3 -m venv ~/st_env } => create vir env
```

```
{ source ~/st_env/bin/activate } => activate
```

Step3: **In Admin**; make sure the requirement.txt file is under “bin” and has permission for all read-write(rw)

Command:

```
{ chmod a+rw /home/admin/ai_env/bin/requirements.txt }
```

Step4: Copy requirements.txt to new user with “bin” directory

Command:

```
{ sudo cp /home/admin/ai_env/bin/requirements.txt /home/student/st_env/. }
```

Step5: login to New User and activate Environment

Command:

```
{ su student } => login
```

```
{ source ~/kusum/bin/activate } => Activate
```

Step6: Install the file using pip inside st_env Environment

Command:

```
{ ~/st_env/bin/pip install -r ~/st_env/bin/requirements.txt }
```

SetUp in GPU

Step1: Linux SetUp

Command:{sudo apt update}

{ sudo apt upgrade }

Step2: Python installation

Command:

{ sudo apt install python }

{ python3 --version }

Step3: Install pip

Command:

{ sudo apt install python3-pip }

Or { sudo apt install python3 python3-pip }

Step4: install Virtual Environment

Command:

{ sudo apt install python3-venv } => install

```
{ python3 -m venv ~/ai_env } => create vir env
```

```
{ source ~/ai_env/bin/activate } => activate
```

Step: install cuda toolkit, we need whole process to install, Cuda Toolkit and Cudnn for the Deep learning projects

Command:

Step4: install PyTorch

Command:

```
{ ~/ai_env/bin/pip install torch torchvision torchaudio }
```

Version check: { pip shows torch }

Step6 : Install Tensorflow/keras/Numpy/Pandas/Matplotlib/Tensorboard/ Seaborn

~~⇒ Use path before installing locally~~

⇒ **Use inside pip (~/ai_env/pip)**

Command:

```
{ ~/ai_env/bin/ pip install matplotlib }
```

```
{~/ai_env/bin/ pip install tensorboard }
```

```
{~/ai_env/bin/ pip install seaborn }
```

Step7: Install jupyter

Command:

```
{ ~/ai_env/bin/pip install jupyter }
```
